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Milestone 1: Theoretical and Technical Post-Mortem

The Prototype as a Piece of Interactive Media:

For this milestone, the goal was to create a monster that had a heavy and threatening presence. This connects to Steve Swink's *Game Feel: A Game Designer's Guide to Virtual Sensation* due to his discussion about a player's perception of game objects and the influence of game rules. The author writes that "The amount of damage something can take provides feedback on the assumed physical properties of the object." (Swink, 183). Although my prototype does not implement the same system, the same notion can be applied to movement, appearance, and game feedback to communicate the character's intended feel.

The character can do simple maneuvers such as stand idle, walk, run, jump, and attack. In the game panel, the character is viewed with a third-person camera that shakes and triggers a sound effect as it executes each footstep. This was done to emphasize the weight of the character because the character alone did not feel menacing enough. To further support the weight of the character, I slowed down the walk and run animation due to the fact most large characters in video games are typically slower. The combination between character appearance, animation speed, and camera manipulation contribute to its physical presence, enhancing the player's experience.

While implementing the camera shake and audio for the footsteps, it was important to be aware of the game response and its impact on player feedback. Swink defines response as "How

the system processes, modulates and responds to player input in real time “ (186). In the case of the shaking camera, excessive shaking made the character difficult to control, interfere with the ability to see the environment, and could potentially cause physical discomfort to the player.

Core Engine Systems:

1. New Input System: The system was used to implement bindings for player controls (WASD, Shift to Run, Space to Jump, and Gamepad).
2. Blend Trees: The system was used to blend animations together so the character did not ‘snap back’ between each animation.
3. Cinemachine: The system was used to create a following, third-person camera. The Impulse Source and Impulse Listening.were used to make the camera shake for each footstep.
4. Animator and Controller: The system was used to control the movement of the character.
5. Audio System: The System was used to implement sound for character footsteps.

Technical Hurdle:

The biggest technical hurdle was scripting in C# for the time. Although scripting as a whole proved to be a challenge, the greatest roadblock I came across was implementing the jump animation in the script. The reason for this was the animation itself has a windup, so when switching between ground only animations, like walking or running, it would either stop the previous animation completely or it would jerk the character before starting the jump animation. To solve this, I watched numerous YouTube tutorials on animation and searched up different scripting methods to switch between animations. I finally came across a method where you set different blend times depending on the animation that is taking place. I set it so it would be a faster blend if the character was jumping or attacking and slower if any other state was active.

This not only was the most technically difficult aspect of the project, but it was also the most time consuming.

Module 5 - Game Loops and Attention Economics

In the book *Advanced Game Design*, it discusses the importance of interactive loops. Interactive loops occur when “The player’s actions are inputs into the game’s loop, and the subsequent change in state in the game model is communicated back to the player, acting as input to the player’s own model and state.” (Sellers, 221-250). While creating my character, I made sure that each animation state would be obvious to the player, ensuring that there is clear communication regarding what is currently happening and what potential next moves are available to them. This forms a healthy interactive game loop for the player, supporting their gaming experience. Additionally, the camera shake adds an additional layer of feedback to the player, indicating that they are heavy and slow, but also, most likely physically strong compared to other game entities.

Additionally, my prototype can be connected back to Evan’s *The Economics of Free* due to the discussion surrounding the economics of attention within the gaming industry. Although my prototype is not a fully functioning game or has any ‘freemium’ aspects to encourage continuous gameplay, interactive loops could contribute to the “attention economy” (Evan, 563-580). Implementing an effective game loop encourages players to continue playing and improve player engagement.

Unity Architecture Compromise

Throughout this post-mortem, I spoke a lot about the shake effect of the cinemachine camera when the character walks, this was not the original idea for my monster character. I originally planned to have a dramatic up and down motion to simulate intense breathing. Due to

Unity's NoiseSettings asset, it was very difficult to accurately create a breathing motion since the exhale of my character was significantly longer than the inhale. No matter how much amplitude I placed in the asset, it either was too subtle or too dramatic that it would not sync properly with the audio. In the end, I decided to scrap the idea and emphasize the character's size rather than try and fill the silence when it is standing idle.

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